C3 - Space shuttle attitude sim

```
w0 = 2;
G = 1;
ts = 0.1;
num = G*w0^2;
den = [1 0 w0^2];
A = [0 1; -w0^2 0];
B = [0 ; G*w0^2];
C = [1 0];
D = 0;


% == transfer function c2d
sysc=tf(num, den);
sysd=c2d(sysc, ts, 'zoh');
[numd, dend] = tfdata(sysd, 'v');

% == state s[ace re[resentation
ssc = ss(A, B, C, D);
ssd = c2d(ssc, ts, 'zoh');
[Atil, Btil, Ctil, Dtil] = ssdata(ssd);

%display
sysd
```

```
sysd =

  0.01993 z + 0.01993
  -------------------
   z^2 - 1.96 z + 1

Sample time: 0.1 seconds
Discrete-time transfer function.
Model Properties
```

```
numd, dend
```

```
numd = 1×3
         0    0.0199    0.0199
dend = 1×3
    1.0000   -1.9601    1.0000
```

```
Atil, Btil, Ctil, Dtil
```

```
Atil = 2×2
    0.9801    0.0993
   -0.3973    0.9801
Btil = 2×1
    0.0199
    0.3973
Ctil = 1×2
     1     0
Dtil = 0
```